WATER RESOURCES, WELLS & PUMPS · NOTE WR05

Canal Systems — Components, Capacities, Outlets and Warabandi

Course topics: IDE8.17 L1, IDE8.17 L9, IDE8.17 L14

The canal system

An irrigation canal system consists of canal works and canal reaches. A canal takes off from the headworks of a storage dam, a barrage or a diversion dam, and the network divides into canals, distributaries and water courses — classified by capacity and by orientation with respect to the headworks.

Component	Description	Capacity (as in the notes)
Main canal	Takes supply directly from the head regulator of a reservoir or diversion structure; size set by its design discharge; field outlets generally not provided	425 to 280 m^3/s
Branch canal	Takes off from the main canal and carries water to major parts of the area; large branches seldom irrigate directly, smaller ones may have outlets	8.5 to 4.0 m^3/s
Major distributary	From branch canals (sometimes small main canals); feeds minors and outlets; has cross regulators to control upstream levels and regulate downstream flow	5.5 to 0.75 m^3/s
Minor distributary (minor)	From major distributaries; supplies outlets directly; water courses take off from it	Usually below 750 L/s
Water course	Under the Northern India Canal and Drainage Act: a channel supplied from a canal but not maintained by the State; runs through common land; maintained by the farmers	30 to 120 L/s
Field channel	Takes off from a water course and carries water to individual fields. In rice areas of southern India it is often absent and water flows field to field, which lowers distribution efficiency	—

Check the main-canal range. 425 m^3/s equals 15,000 cusecs, and the usual textbook range is 15,000 to 1,000 cusecs, i.e. **425 to 28 m^3/s**. The "280" printed in the notes is most likely a typo for 28. Treat all these ranges as indicative — actual canals vary widely.

Ownership: the Government builds, operates and maintains the canals and channels down to the minor distributaries; farmers look after the water courses.

Canal outlets

- Place outlets at suitable points; size each to the area it irrigates.
- Common types in northern India: the **Adjustable Proportionate Module (APM)**, open flume outlet, pipe outlet and siphon outlet.

- The opening of an APM can be adjusted to give the desired discharge. Outlet discharge depends on the water level in the channel and the CCA served.

Warabandi — fixing the turns

1. Supply water through the main water course in "turns" — **warabandi** means fixation of turns.
2. Fix each farmer's turn by the size of his holding within the outlet's CCA. In northern India (Punjab, Haryana, western Uttar Pradesh) turns are weekly.
3. Compute the time per hectare = total minutes in a week ÷ total irrigated area (ha), and allot each holding its share.
4. Include the time needed to fill the water course in the last shareholder's turn.
5. The farmer whose land is nearest the water course gets the first turn; the next turn follows in the succeeding week.

> A week has 7 × 24 × 60 = 10,080 minutes. For an outlet commanding 120 ha, each hectare gets 10,080 ÷ 120 = 84 minutes; a farmer with 3 ha gets 252 minutes (4 h 12 min) every week.

Sources: ICAR eCourse 'Minor Irrigation and Command Area Development' — Dr K. Shanmuga Sundaram (TNAU), reviewed by Dr P. Balakrishnan (UAS Karnataka). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Field outlets are generally NOT provided on

(a) minors
(b) water courses
(c) main canals
(d) major distributaries

Q2. Water courses are maintained by

(a) the State
(b) the farmers
(c) the Centre
(d) the Panchayat engineer

Q3. Capacity of a water course is about

(a) 30–120 L/s
(b) 750 L/s–1 m³/s
(c) 4–8.5 m³/s
(d) 28–425 m³/s

Q4. In warabandi, the first turn goes to the farmer

(a) with the largest holding
(b) whose land is nearest the water course
(c) at the tail end
(d) chosen by lottery

Q5. Outlet whose opening can be adjusted for a desired discharge

(a) pipe outlet
(b) APM outlet
(c) siphon outlet
(d) open flume

Answers

Q1 — (c) main canals
Q2 — (b) the farmers
Q3 — (a) 30–120 L/s
Q4 — (b) whose land is nearest the water course
Q5 — (b) APM outlet